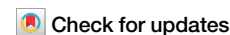


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Thematic cultural heritage tourism trail planning integrating multi-source data and machine learning in Wuhan China



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Nowadays, Cultural heritage tourism faces challenges in route planning, including weak data-mining capacity, limited multi-indicator evaluation, and inefficiencies in traditional pathfinding. This study proposes an innovative thematic and sustainable framework that integrates advanced digital technologies at both meso- and micro-spatial scales to optimize heritage route planning. The study introduces and applies the Non-dominated Sorting Genetic Algorithm III (NSGA-III)—specifically designed for high-dimensional multi-objective optimization—which outperforms existing methods in key aspects and effectively solves complex route optimization problems under multiple constraints. Experimental results confirm that Nsga3ip demonstrating 97% rational route probability and 0.89 optimization efficiency—surpassing MOPSO (83%, 0.62) and random algorithms (12%, 0.19) under identical constraints. The findings demonstrate its strengths in planning quality, enhancement of heritage value, and practicability. This underscores the method's innovation and applicability, further promoting the integration of data-driven approaches in heritage conservation and interdisciplinary urban research.

In recent years, with the rise of sustainable tourism¹, there has been a global focus on the preservation and utilization of cultural heritage². Tourist trails^{3,4}, as a classic form of tourism, have attracted increasing attention in the fields of tourism, urban and rural planning, and architecture since the late 20th century, offering new perspectives on the preservation and utilization of cultural heritage. The concepts of “heritage trail”⁵, “history and culture visiting trail”, and “history and culture walking path” originate from the United States, Korea, and Japan, respectively⁶. It is worth noting that the Chinese extension of these translated concepts is more diverse; Chinese scholars, based on local context, translate them as “history trail”, “heritage tourism trail”, “history and culture visit road”, etc. These concepts refer to examples such as the “Boston Freedom Trail”⁷ in the United States and the “Dr. Sun Yat-sen Historic Trail”⁸ in Hong Kong. These trails physically connect sites via walkways, forming integrated preservation-tourism systems that: (1) chronologically organize distinctive heritage values, (2) coordinate scattered resources with tourism-funded conservation, and (3) enhance experiences through signage, services, and immersive linear spaces for historical exploration. In this study, the concept of ‘heritage tourism trail’⁹ specifically addresses thematic touring systems at urban scales,

forming a conceptual complement to the Council of Europe’s transnational ‘cultural routes’ framework. While cultural routes emphasize cross-regional cultural exchanges, heritage tourism trails focus on thematic integration of local heritage resources through systematic route optimization.

Unlike hastily organized tourist trails that simply connect high-rated attractions, well-planned cultural routes emphasize thematic coherence and narrative continuity, as demonstrated by the European Cultural Routes Program. To support such immersive and layered exploratory experiences, tourism route planning must go beyond superficial criteria and deeply engage with cultural and historical contexts. However, current research in this field often lacks systematic attention to the heritage value of attractions. While some studies begin with tourists’ preferences and behavioral needs^{10,11}, the metrics they use—such as attraction level, popularity, or geographical proximity—tend to prioritize convenience over cultural meaning^{12–14}. As a result, these approaches often overlook key historical background, symbolic significance, or narrative coherence¹⁵, reducing heritage routes to collections of high-ranking points of interest lacking interpretive unity.

In contrast, recent value-based Optimization models—such as those applied to Istanbul’s ancient water-supply heritage—have demonstrated

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that experiential quality can be explicitly aligned with cultural conservation goals¹⁶, suggesting that a more integrated approach is both feasible and necessary. Recent multicriteria frameworks also demonstrate how to balance tourist preferences with cultural value during route assessment¹⁷.

Beyond historical content, another crucial yet under-addressed dimension is the spatial environment surrounding heritage sites. A comprehensive tourist experience requires consideration not only of individual attractions but also of the quality of the spaces in between—including guide systems, spatial interfaces, pedestrian accessibility, and equipment system^{18–20}. However, many existing studies neglect these factors, often defaulting to a two-dimensional selection of attractions without evaluating spatial legibility or environmental coherence. This results in routes that lack guidance, comfort, and engagement. Some recent frameworks have begun incorporating spatial semantics and real-world environmental data into their evaluation systems^{21,22}. However, a gap remains in operationalizing such data within a dynamic, visitor-centered context.

Furthermore, to cope with the increasing scale and complexity of heritage tourism planning, many scholars have adopted Optimization algorithms, particularly single- and multi-objective evolutionary algorithms, to automate decision-making processes^{23,24}. These methods have demonstrated scientific merit, especially in improving planning efficiency when handling large volumes of data across historical urban areas. For example, some researchers have applied the Dijkstra algorithm in ArcGIS to generate optimal routes²⁵. While effective in minimizing time or distance, such methods often overlook experiential richness and thematic depth. Earlier heuristic approaches—such as GA and simulated annealing—proved cost-efficient yet offered limited thematic customization²⁶. Ant Colony Optimization (ACO) and similar algorithms offer another approach^{27,28}, but ACO may encounter issues such as premature convergence, reduced solution diversity, and sensitivity to parameter settings, which can lead to tourist congestion or lack of adaptability in the face of dynamic conditions. Nonetheless, more

recent developments in recommendation systems have begun to integrate cultural narratives with spatial data, offering promising ways to enrich visitor experiences while maintaining algorithmic flexibility²². For instance, Öztürk (2024) combined ACO with 2-Opt to optimize cultural routes, though it required predefined cultural weights. More recent advances have further leveraged optimization-driven recommender systems to fuse spatial context with cultural narratives, demonstrating enhanced personalization for urban heritage tours²¹, though computational costs remain underexplored.

Accordingly, based on a critical review of current research on cultural heritage tourism, this study aims to address three core issues. First, the lack of a comprehensive evaluation framework: existing approaches often fail to systematically integrate heritage value, historical context, and spatial environment—particularly at the micro-scale spatial interfaces—and struggle to accommodate the diverse preferences of tourists. Second, the issue of scale fragmentation: at the meso level, there is an urgent need to overcome the scattered distribution of high-value heritage resources by establishing thematic linkages that activate heritage vitality and promote sustainable reuse. At the micro level, planning must coordinate spatial route experience with the environmental renewal of historic districts. Third, the inefficiency of algorithmic solutions: current methods lack effective tools to resolve the complex multi-objective Optimization challenges posed by cultural heritage tourism route planning in historic cities. To address these gaps, this study proposes an integrated methodology designed to overcome these limitations and ultimately facilitate a transition in heritage conservation from fragmented, “specimen-style” preservation to a more holistic, context-driven spatial narrative approach.

Methods

Methodological framework

The main methodological framework of this study is illustrated in Fig. 1 below.

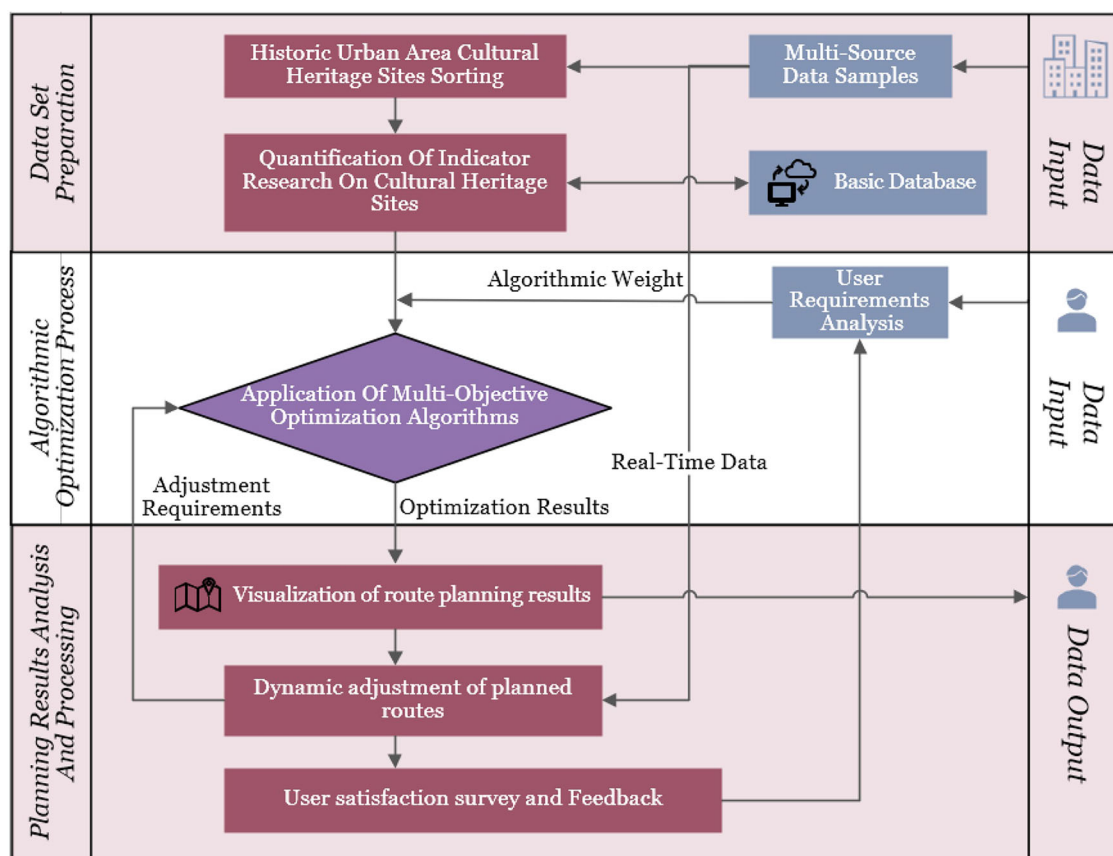


Fig. 1 | Research methodological framework. This figure illustrates the main methodological framework of this study.

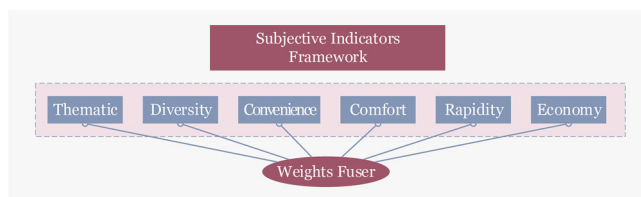


Fig. 2 | Subjective weight assignment.



Fig. 3 | Heritage value evaluation system.

The research framework comprises four sequential stages: (1) Quantification of Indicator Research on Cultural Heritage Sites as input; (2) many-objective algorithmic optimization; (3) route generation, and (4) visualization and output. To ensure the proposed planning method functions effectively in real-time scenarios—and that its outcomes align with both cultural conservation goals and visitor experience enhancement—it is essential to first establish a systematic evaluation framework. This requires identifying a set of relevant indicators that can quantitatively and qualitatively capture the multidimensional attributes of heritage tourism routes. By doing so, the model can translate abstract planning objectives into measurable parameters that support algorithmic optimization. The construction of this indicator system thus serves as a critical foundation for integrating user preferences with spatial and cultural data, enabling personalized and context-sensitive route planning in historic urban environments.

Selection of relevant indicators

Following the definition of the overall planning methodology, the next step involves defining the relevant indicators for evaluating tourist trails, which serve as the basis for assessing the cultural heritage value and fulfilling tourists' needs.

In terms of Subjective Indicators, empirical evidence confirms that tourist preference orientation constitutes the cornerstone of thematic route personalization, where multi-dimensional preference profiling enables precise demand-response planning^{29,30}. Building on this theoretical foundation, this study categorizes travel preferences of distinct user groups into six dimensions: Thematic, Diversity, Convenience, Comfort, Rapidity, and Economy, as illustrated in Fig. 2.

Specifically, Thematicity emphasizes the design of tourism routes centered on particular themes (e.g., history, religion, culture) to enhance tourists' immersive experience; Diversity focuses on providing rich experiences through popular and abundant attractions; Convenience addresses the accessibility of commuting, dining, and accommodation; Comfort reflects the quality of the spatial environment by avoiding high-traffic areas; Rapidity concerns time efficiency during transit; and Economy evaluates cost-effectiveness relative to budget. These six indicators are processed through the Weights Fuser with adjustable weights.

In terms of Objective Indicators, the heritage value assessment framework developed in this study adopts a three-tiered structure that synthesizes authoritative sources: (1) Statutory foundations from China's cultural protection system—namely the *Law on Cultural Relics Protection* (2002 Revision) and the *Standards for Protection of Cultural Relics*^{31,32}; (2) Internationally recognized charters, including the *Venice Charter*³³, the *Nara Document on Authenticity* (1994)³⁴, and the *ICOMOS China Principles* (2015)³⁵; (3) Contextual adaptations, resulting in a four-dimensional evaluation system comprising Historical Value, Aesthetic Value, Scientific Value, and Social Value Figs. 3, 4.

This framework is intentionally designed to align with China's national development priorities while addressing methodological limitations in conventional heritage assessments. Specifically: Historical Value (HV) integrates UNESCO Criterion (vi) to emphasize Wuhan's revolutionary legacy³⁶; Aesthetic Value (AV) incorporates principles of traditional Chinese landscape aesthetics following ICOMOS-IFLA protocols³⁷; Scientific Value (SV) adapts ICCROM material standards to reflect regional architectural techniques³⁸; Social Value (SoV) reflects participatory models grounded in China's ecological civilization strategy³⁹.

The selected four-dimensional system (abbreviated as HV, AV, SV, and SoV) offers a balanced integration of international conservation principles with the distinct characteristics of Chinese urban heritage. This ensures methodological robustness while responding to the unique conservation needs and policy contexts within China.

Thematic Relevance (TR) measures the strength of association between a tourist attraction and predefined thematic routes. High-TR routes systematically integrate thematically aligned attractions, thereby enhancing thematic immersion, educational value, and overall experiential quality⁴⁰. As an objective indicator, TR quantifies the degree of match between an attraction and the theme assigned by the tourist route. This indicator links to the Thematic dimension in the subjective indicator framework, aiming to meet the needs of tourists seeking in-depth thematic experiences.

For example, consider the attraction "Yellow Crane Tower". Assume the predefined themes include the Yellow Crane Tower Historical Trail, Sun Yat-sen Historical Trail, Mao Zedong Historical Trail, and Zhang Zhidong Historical Trail. The method of assigning thematic relevance to the "Yellow Crane Tower" attraction is demonstrated in Table 1 below.

In terms of Spatial Environment, micro-scale built environment characteristics significantly influence pedestrian behaviors, as demonstrated in historic street studies^{41,42}. Building upon this foundation, we adapt a set of spatial environment evaluation indicators from established scholarly literature^{18–20}, including: ①Historical Environment Quality (HQ), ②Landscape Environment Quality (LQ), ③Service Facilities Quality (SQ), ④Guide System Quality (GQ), ⑤Population Density (PD), ⑥Internal Accessibility (IA), ⑦Discussion Heat (DH), ⑧Praise Rate (PR).

Among these, the Historical Environment (HQ) is proposed to be quantified through image analysis and streetscape image analysis, evaluating the historical atmosphere of attractions by assessing architectural style unity and visibility of historical elements. The Landscape Environment (LQ) is calculated statistically using the mean values derived from image analysis, user feedback, and expert ratings, reflecting the degree of greening and the arrangement of buildings. The Service Facilities Quality (SQ) measures convenience and service level by counting the number and type of supporting facilities such as public toilets, catering venues, and rest areas per square kilometer. The Guide System Quality (GQ) evaluation is based on field research and user questionnaires, illustrating how effective signage improves visitors' comprehension of attraction information. Discussion Heat (DH) and Praise Rate (PR) are calculated as weighted averages of comments on various platforms: the former reflects the frequency of attraction-related discussions on social media and travel websites, while the latter indicates overall tourist satisfaction. Internal Accessibility (IA) quantifies transportation efficiency within the scenic area through two core metrics: ①Path network complexity: Analyzed via GIS-based road connectivity indices; ②Pedestrian walkway continuity: assessing unobstructed connections between key functional areas. P Population Density (PD) reflects the distribution and flow of tourists within each area of the attraction. The optimal value for this indicator lies within a reasonable range, as values too high or too low may negatively affect visitor experience.

In terms of Reverse Factors, existing studies demonstrate that constraint factors play a critical role in balancing tourist preferences¹⁰, particularly in reconciling competing objectives such as time efficiency and experience depth. Therefore, this study proposes three reverse factors that must be incorporated during route optimization: ①Dwell Time (DT): The total duration of tourist activities (such as visiting, photographing, and resting) at each attraction, reflecting the intensity of the visitor experience;

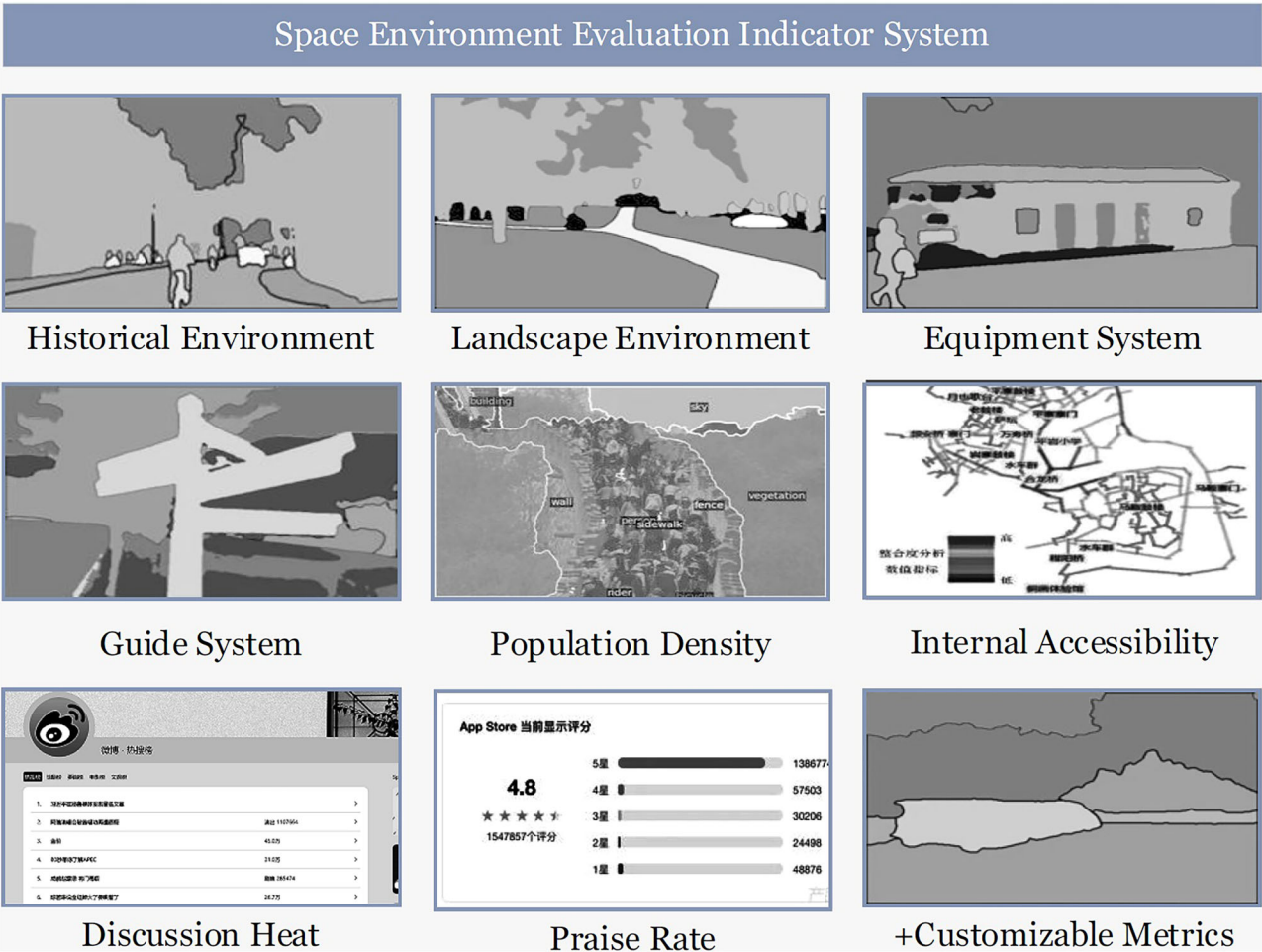


Fig. 4 | Space environment evaluation indicator system.

Table 1 | Thematic Relevance (TR) Assessment for Yellow Crane Tower Across Preset Trails

No.	Thematic Trail	Clarification	Relevance Score
1	Yellow Crane Historic Trail	As the core attraction, directly embodies the history and culture of Yellow Crane Tower	3
2	Wuchang Ancient City History Trail	Closely related to Yellow Crane Tower's historical context, providing rich supplementary information	3
3	Sun Yat-sen Historical Trail	Partial historical linkage with Dr. Sun Yat-sen, though not a primary representation	2
4	Mao Zedong Historical Trail	Moderate correlation with Mao Zedong's historical background, but not a dominant theme	2
5	Zhang Zhidong Historical Trail	Limited historical connections with Zhang Zhidong	1
6	Hanyang Old Town Historic Trail	No relevance to Yellow Crane Tower's history the ancient city of Hanyang	0

The association strength was obtained by inviting historians, cultural heritage experts, etc. to score the association between each thematic trail and the Yellow Crane Tower, using a scoring system from 0 to 3 (3 means strong association, 2 means moderate association, 1 means weak association, and 0 means no relevance). By collecting the scores from all experts and calculating the average value, the final strength of association of each thematic trail was obtained.

② Travel Time (TT): The travel time between consecutive attractions (excluding movement within an attraction), calculated based on actual route distances; ③ Visitor Cost (C): A standardized estimate of expenditure derived from official pricing data and crowd-sourced cost information.

In terms of Analysis of the Correlation Between Subjective and Objective Indicators, the final reference system of indicators is presented in Table 2 below. Furthermore, this section analyses the relationships between subjective and objective indicators to clarify the influence each indicator (as listed in the previous section) exerts on the overall objective function.

To systematically investigate the relationships between subjective and objective indicators, a structured questionnaire was administered to 127 participants. Participants were selected via stratified sampling to ensure a

95% confidence level with a $\pm 5\%$ margin of error. The sample encompassed diverse age groups (18–25: 32%; 26–40: 41%; 41–60: 22%; 60 + : 5%) and visitor types (first-time visitors: 58%; repeat visitors: 42%). All respondents met the eligibility criterion of having visited at least two cultural heritage attractions within the study area in the past year. Demographic data indicated that 44% of participants were male and 56% female; 76% held a bachelor's degree or higher; and 84% were domestic tourists.

The questionnaire employed a 7-point Likert scale to assess bidirectional correlations between subjective and objective indicators, where +3 represented a strong positive correlation, 0 indicated no correlation, and –3 denoted a strong negative correlation. To ensure data reliability, the questionnaire achieved a Cronbach's alpha of 0.81 across all indicator items.

Attention-check questions and response-time monitoring were implemented to identify and exclude invalid responses. The resulting correlation patterns are visualized in Fig. 5.

Description: This heat map illustrates the patterns of relevance between indicators. Higher positive correlations (dark blue) signify stronger relationships between the two indicators, offering valuable insights for further

analysis and research. Conversely, negative correlations (dark red) indicate an inverse relationship. Lighter colored areas suggest that the subjective and objective indicators are not significantly correlated from this perspective Figs. 6, 7.

Representing the above figure in matrix form, R is substituted into the calculation in the later section:

$$R = \begin{bmatrix} 3 & 2 & 2 & 0 & 1 & 1 & 3 & 3 & 0 \\ 3 & 1 & 3 & 2 & 3 & -1 & 1 & 3 & 0 \\ 3 & 1 & 3 & 2 & 3 & -3 & 1 & 3 & 0 \end{bmatrix} \quad (2.1)$$

Notably, the framework is designed to establish direct mappings between three subjective dimensions (Thematic, Rapidity, Economics) and objective indicators, thereby eliminating the need for questionnaire-based quantification. Specifically, thematic evaluation corresponds directly to the Thematic Relevance (TR) metric, where each attraction's TR score (0–3) determines its thematic rating. Rapidity is algorithmically derived from the sum of Dwell Time (DT) and Travel Time (TT) between sites, while Economy is calculated directly from Visitor Cost (C) data. These predefined mathematical relationships—Thematic = TR, Rapidity = DT + TT, and Economy = C—are encoded as deterministic rules within the NSGA-III optimization model, rather than treated as subjective perceptions. Consequently, these dimensions require no survey validation, as their values are generated programmatically from objective datasets rather than self-reported by users. Based on the relevance matrix above, the weights of the Optimization objectives in the many-objective Optimization model can be adjusted under different subjective demand dimensions.

Planning model construction

In terms of Introduction of the Algorithm, the Genetic Algorithm (GA) is the algorithmic framework employed in this study. It is a stochastic search algorithm inspired by the biological principle of “survival of the fittest.” As a classical global optimization method, GA offers high computational efficiency, strong practicality, and robustness, and is well suited to parallel implementation. NSGA (NSGA-III) is a variant of GA specifically enhanced for multi-objective optimization problems, capable of effectively handling multiple conflicting objectives and generating a set of Pareto-optimal (A Pareto-optimal solution refers to a solution obtained during the multi-objective optimization process that cannot be further improved without compromising the performance of at least one other objective) solutions.

Accordingly, this study selects NSGA-III as the core optimization algorithm. Unlike NSGA-II, NSGA-III performs selection by comparing solutions with reference points, thereby improving the uniform distribution of solutions. By introducing a reference point mechanism, it ensures that generated solutions are evenly spread across the objective space, preventing clustering and offering diverse route options. The algorithm is highly adaptable and performs well in high-dimensional objective spaces, making it especially suitable for “many-objective” (In NSGA-II, “multi-objective” typically refers to problems with 2 or 3 objectives, while “many-objective” denotes problems with at least 4 objectives. In this study, the number of optimization objectives is 6) optimization tasks.

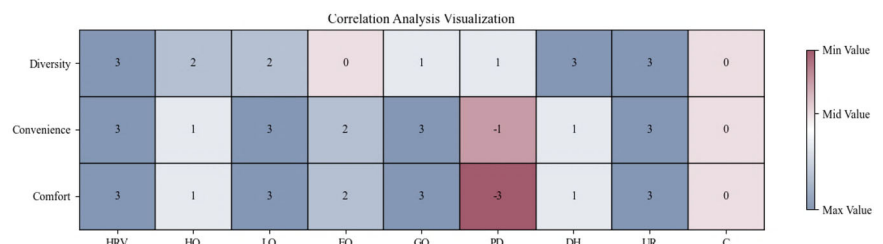
Table 2 | Reference indicator system and relevance analysis for the optimization of heritage tourism trail paths

No.	Evaluation indicators		Relevance
1	Subjective indicators	Thematic	/
2		Diversity	
3		Convenience	
4		Comfort	
5		Rapidity	
6		Economics	
7	Basic information	Name	bound to thematicity
8		Location	
9		Thematic Relevance (TR)	
10	Heritage Value (HRV)	Historical Value (HV)	full positive correlation
11		Aesthetic Value (AV)	
12		Scientific Value (SV)	
13		Social Value (SoV)	
14	Historic Environment Quality (HQ)		
15	Landscape Environmental Quality (LQ)		
16	Equipment System Quality (EQ)		
17	Guide System Quality (GQ)		
18	Population Density (PD)		positively related to diversity, negatively related to comfort
19	Discussion Heat (DH)		full positive correlation
20	Praise Rate (PR)		
21	Internal Accessibility (IA)		positively associated with convenience
22	Dwell Time (DT)		negatively correlated with rapidity
23	Travel time between attractions (TT)		
24	Visitor Cost (C)		negative correlation with economy

Description: 1. Thematic Relevance (TR) measures the degree of correspondence between an attraction and the theme designated for the tour route, and is directly linked to the Thematic dimension; 2. Heritage Value (HRV), encompassing historical, aesthetic, scientific, and social values, is evaluated through expert assessments and standardizations, showing positive correlations with each subjective indicator; 3. Positive factors—including historic environment ambience, landscape environment quality, supporting facilities, signage system, internal footfall intensity, discussion heat, favorable rating, and internal integration—are assessed using various methods and data sources. Some of these indicators exhibit positive or negative correlations with specific subjective indicators.

This item is set with reference to the association strength in Table 1.

Fig. 5 | Relevance matrix for primary and objective indicators.



This flexibility grants NSGA-III a distinct advantage in tourist route planning problems. For example, when tourists place greater emphasis on thematic relevance, the algorithm can adjust weights to priorities attractions with higher thematic association; conversely, when tourists focus more on economic factors, the algorithm favors attractions with lower costs. Thus, the adaptability of NSGA-III allows it to flexibly respond to varying optimization goals under different weight settings.

Furthermore, empirical studies report that NSGA-III achieves superior Pareto diversity when optimizing routes under real-time traffic and environmental inputs. While conventional single-objective optimization algorithms often struggle to satisfy users' multidimensional needs simultaneously^{43,44}, NSGA-III has been widely applied across various route-

planning domains⁴⁵, providing both a solid theoretical foundation and practical experience for this research.

In terms of Model Construction, to better address real-world heritage tourism route planning challenges and enhance the practicality of solutions, this study introduces two major modifications to the baseline NSGA-III framework.

The first is Flexible Constraint Handling in Fitness Evaluation. A dynamic penalty-based constraint mechanism inspired by Ruiz-Meza et al.¹⁰ is embedded within the fitness function to account for budgetary and time restrictions. When the total cost or travel time of the selected set of attractions exceeds predefined thresholds, the corresponding solution incurs a significant fitness penalty. This mechanism effectively filters out infeasible solutions during the evolutionary process, thereby improving the practical feasibility of the generated route plans.

The second is Integration of Dynamic Programming for Path Optimization. The specific steps are as follows (see Appendix for detailed pseudo-code): ① Distance Matrix Computation: Calculate the Euclidean distance between each pair of selected attractions to form a symmetric distance matrix; ② DP Table Initialization: Define the dynamic programming (DP) table $D[\text{mask}][\text{last}]$ following Blank & Deb's reference-point methodology⁴⁵, where mask represents the set of visited attractions to maintain solution diversity and avoid premature convergence. Initialize $D[1][0] = 0$, representing the starting point at the first attraction; ③ Recursive Table Update: Iterate through all valid states of mask, updating $D[\text{mask}][\text{last}]$ by considering transitions from previously visited nodes to the next unvisited node

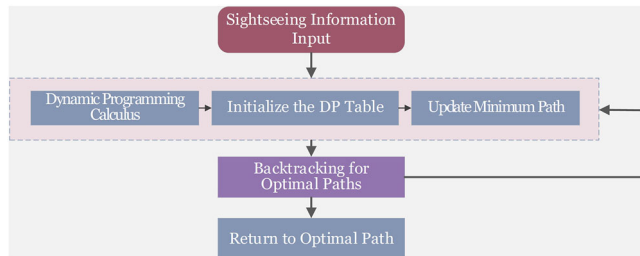


Fig. 6 | Dynamic planning methodology framework.

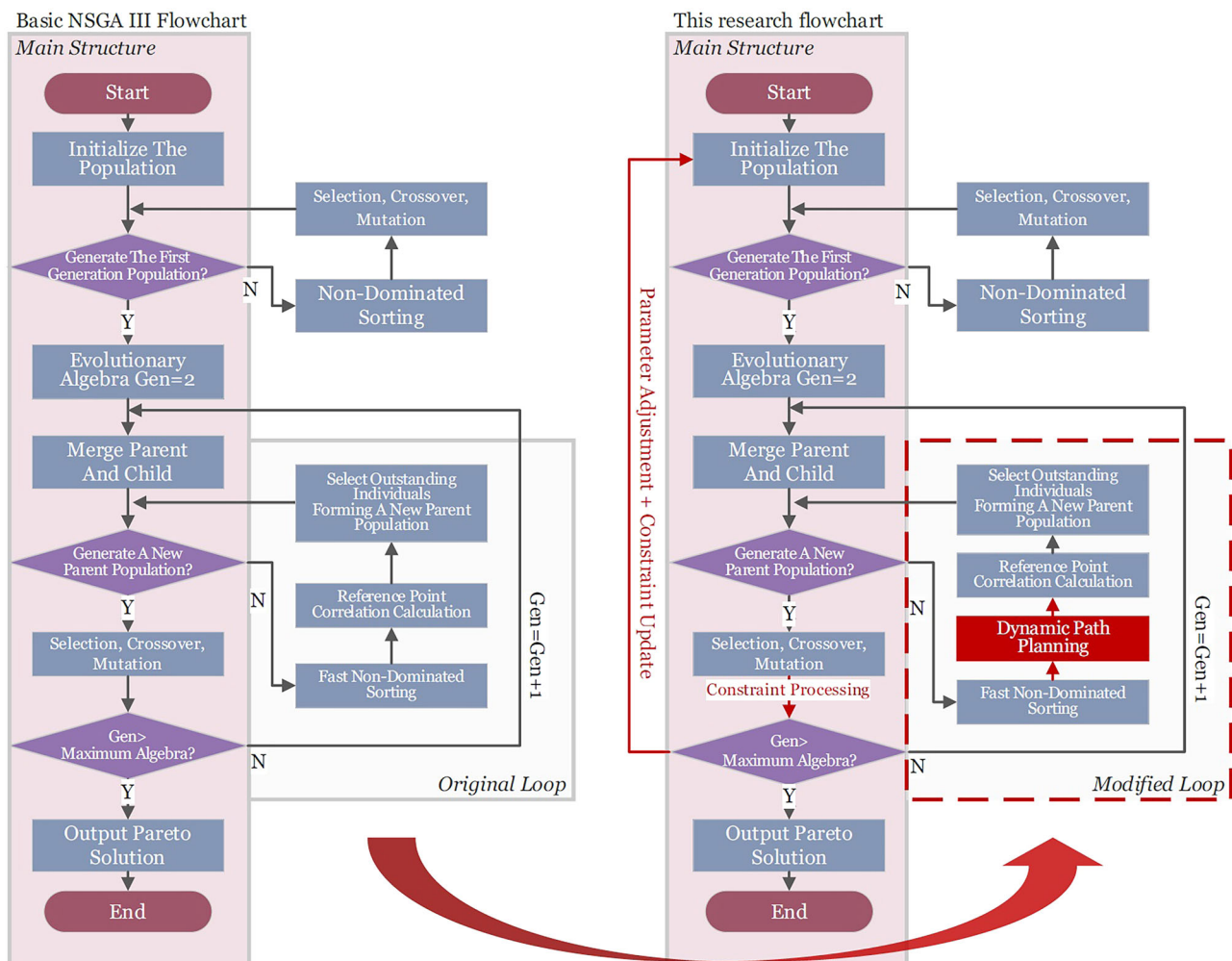


Fig. 7 | Comparison between the Proposed Algorithm and the Baseline NSGA-III Algorithm.

Fig. 8 | Algorithm framework.

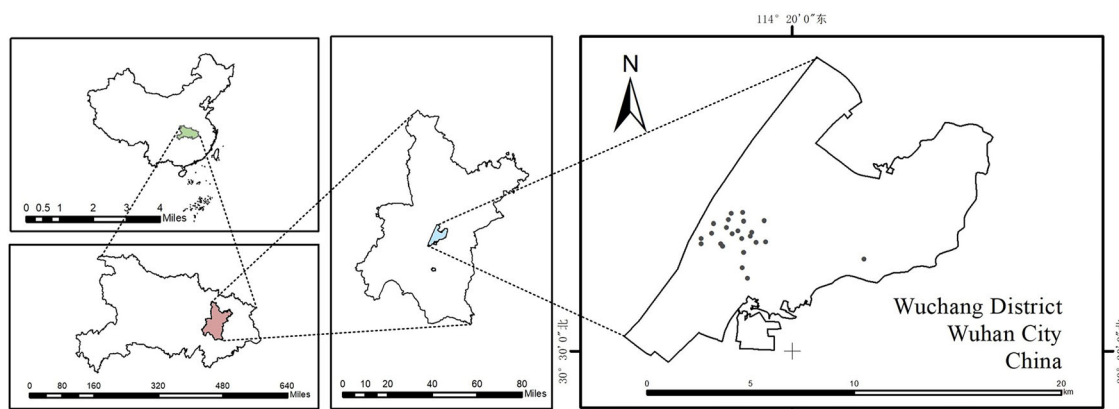
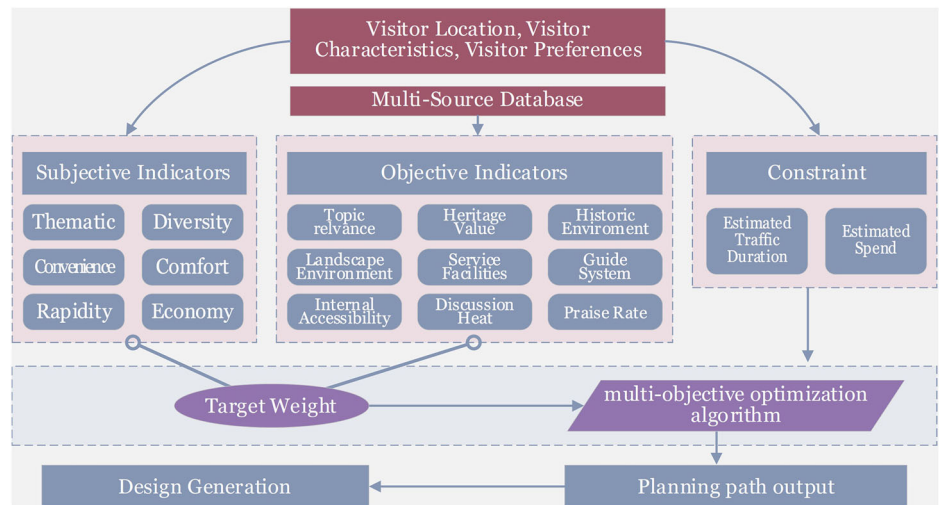


Fig. 9 | Study area and sites.

with minimal cumulative distance; ④ Optimal Sequence Recovery: Perform reverse backtracking from the final state to reconstruct the optimal visitation sequence. Iteratively remove the most recently visited node based on $D[\text{mask}]$ [last] values until the complete route is recovered.

These enhancements to NSGA-III ensure that the generated route solutions are not only Pareto-optimal across multiple objectives but also practical and executable.

Definition of Indicators and Algorithmic Framework: According to Table 1, six Optimization objectives are identified: ① Maximize Thematic; ② Maximize Diversity; ③ Maximize Convenience; ④ Maximize Comfort; ⑤ Minimize Rapidity; ⑥ Minimize Economics. The constraints are defined as follows: ① Total Travel Time limit (Max_Time); ② Total Cost limit (Max_Cost).

Figure 8 illustrates the structure of the many-objective Optimization model for heritage tourism trail planning, highlighting key input and output elements alongside core evaluation indicators.

Following the logic of the NSGA-III model, input variables include thematic-related historical and cultural resource types, geographic coordinates of attractions, site-level environmental conditions, user-generated feedback, and transportation conditions. Based on individual user characteristics and preferences, the relative weights of evaluation indicators are dynamically adjusted to generate optimal heritage tourism routes that cater to user-specific requirements.

Experimental process

In terms of the collecting and organizing background information, the selection of the study area is conducted in three steps. First, the scope of the

historic urban area is defined by consulting relevant sources such as historical archives, local records, and academic research, in order to delineate the boundaries of the heritage site. Second, urban multimodal data are integrated to collect geographic coordinates of buildings. Finally, all collected data are input into a GIS platform to ensure the geographic information is accurate and complete, thereby establishing a foundational information database of the cultural heritage site's contours and key locations.

Systematic organization of information related to the cultural heritage of the historic urban area proceeds as follows. First, the architectural heritage within the historic urban area is organized based on historical records, considering factors such as the age of buildings, significant historical events, and notable figures. Second, this information is systematized and categorized according to different themes—classifying architectural heritage based on criteria such as construction era, event relevance, and residences of prominent individuals—to form diversified historic district tourism trails. Based on extensive field research and expert consultations, and with reference to the cultural heritage tourism planning documents of Wuhan⁴⁶, a preliminary design of thematic trails in Wuhan is presented in the table below Fig. 9.

In terms of the multi-source data extraction based on Wuhan database, this study takes three kinds of data Table 3.

The first is static data acquisition. The analysis of multi-source static urban data facilitates the identification of heritage route segments with optimal cultural resource density and route planning efficiency, forming the foundational basis for enhancing the Optimization of the visitor experience. The specific data indicators and corresponding data acquisition mechanisms are summarized in Table 4.

Table 3 | Thematic conceptualizations for designing tourism trails in historic urban areas (Wuhan as an Example)

Subject Matter	Thematic Trails	Visiting Nodes	Background Information
Event	Historical Trail of the 1911 Revolution	Uprising Gate, Ziyang Lake Park, the former site of the Engineering Camp, the Xinhai Revolutionary Museum, Drum Tower Cave, Li Yuanhong Mansion	The trail shows the historical scenes and events of the 1911 Xinhai Revolution. This revolution overthrew the Qing Dynasty and ended more than 2000 years of imperial rule in China.
	Red Memory History Trail	Ruins of the Agricultural Institute, the site of the Fifth Congress, Mao Zedong's former residence, Tube Lane, the former site of the Central Military and Political School in Wuhan, and the former site of the Office of the Central Military Commission of the Communist Party of China.	Focusing mainly on the important historical activities of the Chinese Communist Party in Wuhan, the trail showcases key events and leaders' activities during the early revolutionary period.
Typology	Historic Trail of Hanyang Ancient City	Guiyuan Temple, Hanyang Catholic Church, Gongyuan Pavilion	The trail showcases Hanyang's historical heritage as an ancient city, including the diverse remains of religion, education and culture.
	Wuhan Industrial Heritage Trail	Dabie Mountain Cliffs, Hanyang-made Creative Park, Guqintai, Qintai Concert Hall, Qintai Square, Hanyang Steel Industrial Heritage	The trail showcases Wuhan's history as a pioneer of modern industrialization in China.
	Wuchang Ancient City History Trail	Yangtze River Bridge, Hanyang Gate, Hubu Lane, the former site of Jingxin Academy, the former site of St Joseph's Church, Zhaiyage, Tanhualin, Wushengmen Ruins	The trail showcases the rich historical heritage of the ancient city of Wuchang, both as an important military fortress and as a center of culture and education.
Famous Buildings, People	Yellow Crane Tower Historic Trail	Monument to the completion of the Yangtze River Bridge, Shengxiang Pagoda, Yellow Crane Tower, Drum Tower Cave, Red Tower, the former site of the Hubei Provincial Library, Longhua Temple, Wuhan University	The Yellow Crane Tower is one of the most representative historical buildings in Wuhan, and the path along the way showcases historical buildings and cultural sites associated with the Yellow Crane Tower since ancient times.
	Sun Yat-sen Historical Trail	Wuchang Uprising Military Government, Wuhan Ejun Dudu Government Site, Shouyi Park, Yellow Crane Tower, Sanmin Road, Zhongshan Park in Hankou, Zhongshan Road	Sun Yat-sen was a pioneer of China's democratic revolution and the trail shows his footsteps in Wuhan.
	Mao Zedong Historical Trail	Yangtze River Beach, the former site of the 1987 Conference, Dazimen Railway Station, Wuchang Central Agricultural Institute, Mao Zedong's former residence, East Lake Hotel, the former site of Wuhan Heavy Machine Tool Factory	The trail documents Mao's revolutionary activities in Wuhan, particularly the early peasant movement and the sites where revolutionary strategies were developed.
	Zhang Zhidong Historical Trail	Hanyang Iron Factory, Hubei Gun Factory, Hubei Weaving Factory, Nantong Dasheng Yarn Factory, Silk Reeling Bureau, the former site of Hemp Bureau, Model Prison, the former site of Beijing-Han Railway, Zhang Zhidong and Wuhan Museum, Zhang Zhidong Memorial Hall, Wuhan University, the former site of the new-style school.	The trail showcases Zhang Zhidong's drive for modernization, including industrial heritage such as the Hanyang Iron Works and educational facilities such as Wuhan University.

Table 4 | Static data collection list

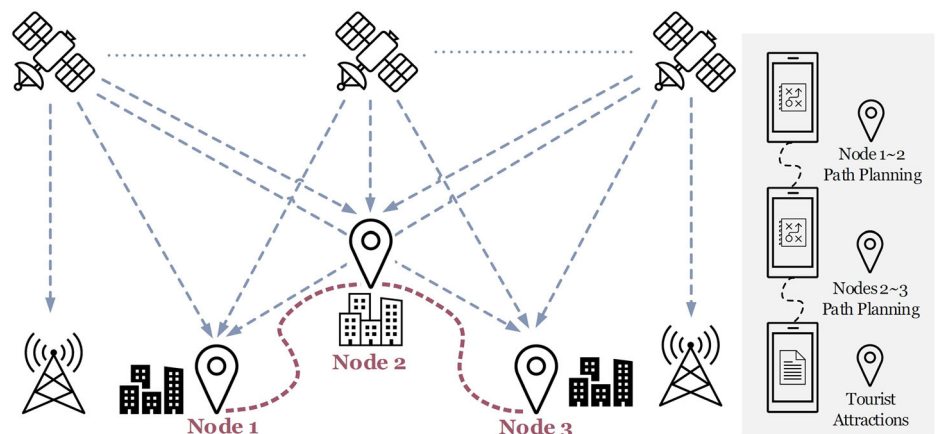
Evaluation indicators	Evaluation basis, criteria	
Basic information	Name	-
	Location	Baidu map API (latitude and longitude) imported into GIS system ⁵²
	TR	The relevance to each of the Thematic presets is taken through a preset value (Refer to Table 1 Settings)
Heritage value (HRV)	HV	Standardized historical significance scores, based on expert evaluation and historical document analysis ⁵³
	AV	0-5 scale based on Likert scales
	SV	
	SoV	
Positive factor	HQ	Historical ambience index after quantitative image analysis, combined with standardized scores after street scene image analysis ¹⁵
	LQ	Composite landscape scores, statistically calculated by averaging image analysis, user feedback and expert ratings ^{18,54}
	EQ	Facility density index, counting the number and type of supporting facilities per square kilometer ⁵⁵
	GQ	Marking clarity index, statistically measured by the mean value of user questionnaires and expert ratings ⁵⁶
	PD	Baidu Heat Map (Heat Zone) ²²
	DH	Heat statistics, through the weighted average of the heat of each platform for statistics ⁵⁷
	PR	Average positive review rate, aggregated and weighted from multi-platform user evaluations ⁵⁸
	IA	Accessibility score based on internal path connectivity, collected via field surveys or spatial syntax analysis ¹⁹
Reverse factor	TT	Travel time index calculated as weighted average of path length and assumed walking speed between attractions ²²
	C	Median visitor cost, normalized from crowd-sourced and official pricing datasets across platforms

The second is dynamic data acquisition. During the user's tour, the system dynamically adjusts and updates the tour route in real time based on information from attractions, city traffic data⁴⁷, and weather conditions⁴⁸. For example, if the user deviates from the established route, the system will prompt them to adjust their direction or skip nodes that are too distant. In

the event of sudden weather changes, the system advises appropriate actions, such as purchasing rain gear or opting to visit indoor attractions. If congestion occurs around a specific attraction, the system recommends alternative transport modes. The indicators monitored in real time on the platform, along with their descriptions, are presented in Table 5 below.

Table 5 | Dynamic data collection list

No.	Indicator	Description of indicators	Data acquisition
1	Weather change	Visitors are outdoors and weather can have a significant impact on the experience	Weather APIs provided by the Met Office
2	Traffic condition	Avoiding tourist disruption due to traffic problems	Traffic information interface of major map service providers
3	Attraction Operation	Avoid visitors missing the opening hours of attractions	The official website of the attraction, the data interface of the Culture and Tourism Bureau, and the interface provided by tourism information aggregation platforms
4	Volume and congestion	Avoid overcrowded attractions or areas to provide a better experience.	GIS data, data interfaces provided by specific attraction managers, third-party data providers
5	User preferences and real-time feedback	Adjust routes based on feedback from visitor demand experiences to provide trails that are more in line with user tastes.	User Interactive Data Interface (UIDI) within the system
6	Emergency or security alerts	In the event of an emergency or security alert, the ability to adjust routes or provide security advice in a timely manner ensures safety.	Public safety data from government announcements, police or fire stations
7	Special events or festivals	Stay informed about special events or festivals in real time and consider whether you need to adjust your route to participate in or avoid them.	Information provided by government departments of culture and tourism, organizers of specific events

Fig. 10 | Location information extraction mechanism.

At the current stage, all static data indicators listed in Table 4 have been integrated and applied in the computational experiments conducted in this study. However, the dynamic data adjustment system described above remains conceptual and has not yet been implemented or empirically tested. Its design represents a prospective enhancement for future platform development. No real-time simulation or field deployment of the dynamic module has been performed within the scope of the present study.

The third is the user feature data extraction, which are derived from individual behavioral patterns and activity history. When such behavioral data are insufficient, users are encouraged to manually provide preference information prior to departure. Before executing the routing algorithm, the system assesses the volume of the target user's tourism history data: ① if the data volume exceeds a predefined threshold, a preference-based recommendation algorithm is applied to generate personalized travel tags for the user; ② if the data volume is equal to or less than the threshold, the system employs a default (equal-weight) routing recommendation algorithm to suggest attractions.

User locations are obtained in real time via GPS-based mobile device positioning services. This serves as the basis for defining the starting point of the tour route and enables dynamic adjustment during the tour.

While the user is on the move, the system continuously monitors their location in order to adapt the navigation path and attraction information based on real-time traffic conditions, timing constraints, and user feedback. The mechanism for collecting and processing user location data is illustrated in Fig. 10 below.

During the tour, if the system detects that the attraction at the user's next stop has been closed within the expected tour time, the system will suggest the user to skip the node and go to the next one directly; if the user's visiting duration at a certain attraction is too long, the system will carry out a reminder to speed up the tour progress in the attractions in the back or skip certain attractions that have a relatively low recommendation value.

Additionally, upon completion of the tour, users are invited to complete a satisfaction questionnaire covering dimensions such as landscape, transport, and service experience^{49–51}. These assessments are utilized to improve the system and will be incorporated into subsequent iterations of the Thematic Route Recommendation Mechanism.

In terms of the data processing, this study compiled twelve evaluation indicators relevant to cultural heritage attractions within the historical city of Wuhan, including Coordinate Position (CO), Thematic Relevance (TR), a set of attraction evaluation indicators (HRV through DT), as well as Travel Time (TT) and Visitor Cost (C).

Due to the extensive data volume, analysis focused on 25 representative cultural heritage attractions located in the Wuchang Historic District. Their corresponding static data were incorporated into the planning experiments, serving as empirical input for the improved NSGA-III algorithm.

For the data set processing, the dataset utilized in this study is presented in Table 6 below. It was normalized (All indicator values were normalized to a [0, 1] scale using min-max normalization to ensure comparability across metrics and prevent scale bias during multi-objective optimization) for use in the experiments (see Appendix for the full table contents).

Table 6 | Experimental data set settings

ID	TR	HRV	HQ	LQ	EQ	GQ	PD	DH	PR	IA	TT	C	CO (X)	CO (Y)
1	0.6	0.8	0.3	0.7	1.0	1.0	0.3	0.0	0.5	1.0	0.0	0.0	114.3005	30.5520
2	0.8	0.8	0.7	1.0	1.0	1.0	0.3	0.3	0.5	1.0	0.2	0.3	114.3074	30.5507
3	1.0	1.0	1.0	1.0	1.0	1.0	0.7	1.0	1.0	1.0	1.0	1.0	114.3090	30.5503
4	0.6	0.8	0.7	0.3	0.5	0.5	0.3	0.0	0.0	0.5	0.0	0.3	114.3152	30.5498
5	0.8	0.8	0.7	0.7	0.5	1.0	0.3	0.0	0.5	0.5	0.3	0.0	114.3177	30.5470
6	0.8	0.8	1.0	0.7	0.5	1.0	0.3	0.3	0.5	0.5	0.0	0.3	114.3219	30.5472
7	1.0	1.0	1.0	1.0	1.0	1.0	0.7	1.0	1.0	0.0	1.0	0.0	114.3727	30.5428
8	0.8	0.8	0.7	1.0	1.0	0.5	0.7	0.7	0.5	1.0	0.0	0.0	114.3010	30.5564
9	0.8	1.0	1.0	0.7	0.5	1.0	0.7	0.3	0.5	0.5	0.0	0.0	114.3003	30.5540
10	0.8	0.4	1.0	0.7	0.5	1.0	1.0	1.0	1.0	0.5	0.3	0.8	114.3050	30.5536
11	0.4	0.8	0.7	0.7	0.5	0.5	0.3	0.3	0.5	0.0	0.0	0.3	114.4175	30.5522
12	0.2	0.8	0.7	0.3	0.5	0.5	0.3	0.3	0.5	0.0	0.0	0.2	114.3124	30.5562
13	0.4	0.8	0.7	0.3	0.5	0.5	0.3	0.0	0.5	0.0	0.0	0.0	114.3118	30.5596
14	0.4	0.8	0.7	0.7	1.0	0.5	0.7	1.0	0.5	0.5	0.5	0.8	114.3212	30.5563
15	0.0	0.8	0.7	0.3	0.5	0.5	0.3	0.3	0.5	0.5	0.3	0.3	114.3070	30.5594
16	0.0	0.8	0.7	0.0	0.5	1.0	0.3	0.0	0.5	0.5	0.3	0.2	114.3058	30.5567
17	0.0	1.0	0.7	0.0	0.5	1.0	0.3	0.0	0.5	0.5	0.5	0.0	114.3051	30.5566
18	0.2	1.0	0.7	0.3	0.5	1.0	0.0	0.0	0.5	0.5	0.3	0.3	114.3033	30.5453
19	0.2	0.8	0.3	0.0	0.5	1.0	0.3	0.0	0.5	0.5	0.3	0.2	114.3033	30.5453
20	0.2	0.8	0.3	0.3	0.5	1.0	0.3	0.0	0.5	0.5	0.0	0.2	114.3159	30.5513
21	0.0	0.2	0.0	0.3	1.0	0.5	0.7	0.3	0.0	0.5	0.3	0.8	114.3122	30.5448
22	0.4	0.8	0.7	0.3	1.0	1.0	0.3	0.3	1.0	0.5	0.5	0.0	114.3124	30.5427
23	0.0	0.6	0.3	0.0	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	114.3145	30.5388
24	0.0	0.0	0.0	0.7	0.5	0.5	0.7	0.3	0.5	0.5	0.5	0.0	114.3118	30.5361
25	0.4	0.6	0.3	0.3	0.0	0.0	0.3	0.0	0.0	0.5	0.0	0.0	114.3139	30.5314

Description: The Thematic Relevance (TR) values in this table depend on the relevance of each attraction to the thematic 'Yellow Crane Tower Historic Trail'.

For the optimization objective and constraint setting, firstly, the objective function is modeled (In a many-objective optimization model, there are usually two or more conflicting objectives, and the goal of optimization is to find an equilibrium such that each objective is satisfied to some extent) based on the subjective indicators listed in Table 2, as follows:

$$\max \text{imize} \begin{cases} \text{thematic} = \sum TR_i \\ \text{diversity} = \sum_{j=1}^n \omega_{D,j} \cdot \sum_{i=1}^m f_{D,j,i} \\ \text{convenience} = \sum_{j=1}^n \omega_{C,j} \cdot \sum_{i=1}^m f_{C,j,i} \\ \text{comfort} = \sum_{j=1}^n \omega_{F,j} \cdot \sum_{i=1}^m f_{F,j,i} \end{cases} \quad (3.1)$$

$$\min \text{imize} \begin{cases} \text{travel_time} = \sum T_i + \sum TT_i \\ \text{economy} = \sum C_i \end{cases} \quad (3.2)$$

where ω is the set weight of the value indicator (The set weights of the value indicators are the relative importance assigned to the different optimization objectives or evaluation indicators in many-objective optimization. These weights reflect the extent to which each underlying indicator influences the final outcome during the optimization process), respectively, where the data are derived from the matrix R (Eq. 2.1).

Secondly, the optimization process must comply with the following constraints to ensure the planning scheme aligns with the user's basic situation and needs: ① Time constraint: The total visit time must not exceed the user-defined time limit. This includes the visit time at each attraction and

the commuting time between attractions to guarantee completion of the tour within the specified period; ② Budget constraint: The total cost must not exceed the user's budget. This requires the rational selection of attractions during route planning to control overall expenses. Accordingly, the time constraint (Max_Time) is set as:

$$\sum_{i=1}^n T_i + \sum_{i=1}^n TT_{i-1} \leq \text{Max_Time} \quad (3.3)$$

The budget constraint (Max_Cost) is set as:

$$\sum_{i=1}^n C_i \leq \text{Max_Cost} \quad (3.4)$$

In terms of the NSGA-III Parameter Setting, to ensure the trail planning scheme meets users' actual needs, six optimization objectives are set, with initial objective weights uniformly assigned to allow subsequent adjustment according to user preferences.

Additionally, the algorithm's maximum number of iterations is set to 150 generations, with an elite proportion retaining 50% of individuals; the crossover probability is set to 0.5, and the mutation probability to 0.2 to enhance population diversity; the maximum number of generations without improvement is set to 200 as a convergence criterion. Detailed parameter settings are shown below.

In terms of the Optimization Process, after generating the initial population, the following steps are performed: ① Crossover operation: Crossovers are performed on parent individuals to generate offspring and increase population diversity; ② Mutation operation: Mutation is applied to

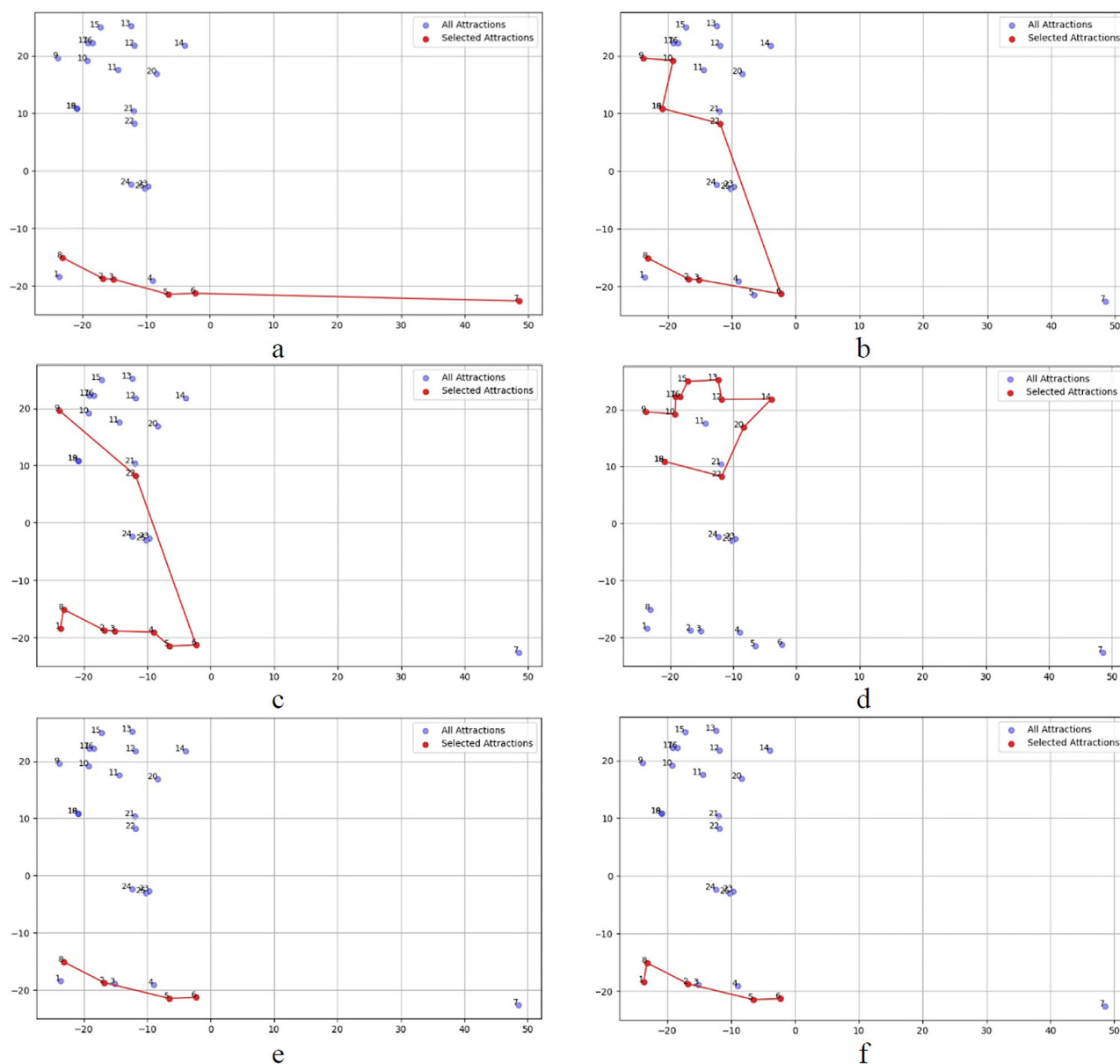


Fig. 11 | The results of the route planning generated by the algorithm. These figures shows high weight planning results of theme (a), diversity (b), convenience (c), comfort (d), speed (e), and economy (f).

some individuals to further enhance diversity; ③ Non-dominated sorting: Parent and offspring populations are merged and sorted to identify the Pareto frontier; ④ Introduction of reference points: A set of reference points (or ideal points) is generated based on the number of objective functions to ensure uniform distribution of solutions; ⑤ Elite retention: The highest-scoring individuals in the current population are retained to ensure transmission of high-quality solutions; ⑥ Output of results: Ultimately, the highest scoring individual from the population is selected to output the optimal travel route, including route, total time, total cost and score.

Results

Planning results and algorithm performance

The route planning results generated by the algorithm are shown in Fig. 11. The algorithm demonstrated a high degree of scientific rigor and practical applicability in many-objective Optimization. Under the Thematic Relevance (TR) objective, it selected relatively longer routes that included attractions with the highest thematic relevance scores (e.g., Points 2, 3, 5, 6, 7, and 8), adhering to the principle of theme-first planning. For the objectives of Diversity and Convenience, the

algorithm prioritized attractions exhibiting superior historical and environmental qualities under varying weighting conditions (as defined by matrix R). It also favored indicators with high popularity or user approval, such as Heritage Value (HRV), Historic Environment Quality (HQ), Landscape Environment Quality (LQ), Equipment System Quality (EQ), Guide System Quality (GQ), Discussion Heat (DH), and Praise Rate (PR). Under the Comfort objective, the algorithm focused on attractions scoring highly for comfort-related factors, characterized by elevated EQ and GQ values alongside low Population Density (PD). For the Rapidity objective, it tended to select attractions with shorter visit durations and shorter travel distances, reflected in lower Travel Time (TT) values. Under the Economics objective, the algorithm aimed to minimize Visitor Cost (C), thereby ensuring cost-efficiency. Finally, when subject to strict constraints (Max_Time < 60 minutes, Max_Cost < 100 RMB), the algorithm selected a smaller subset of attractions with relatively high overall scores. This outcome aligns with conventional planning principles—balancing constraint compliance with visit quality—and underscores the algorithm's flexibility and adaptability in real-world scenarios.

Fig. 12 | Convergence analysis of the algorithm under various Optimization objectives.

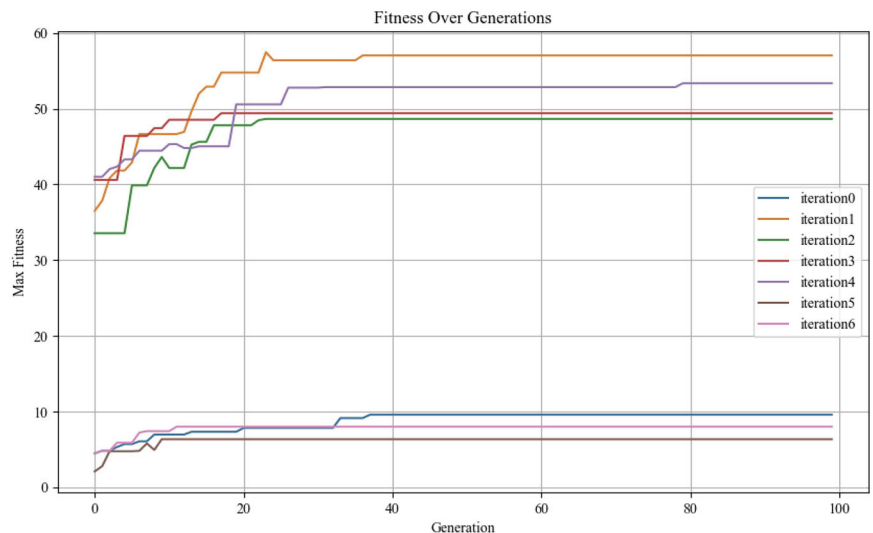
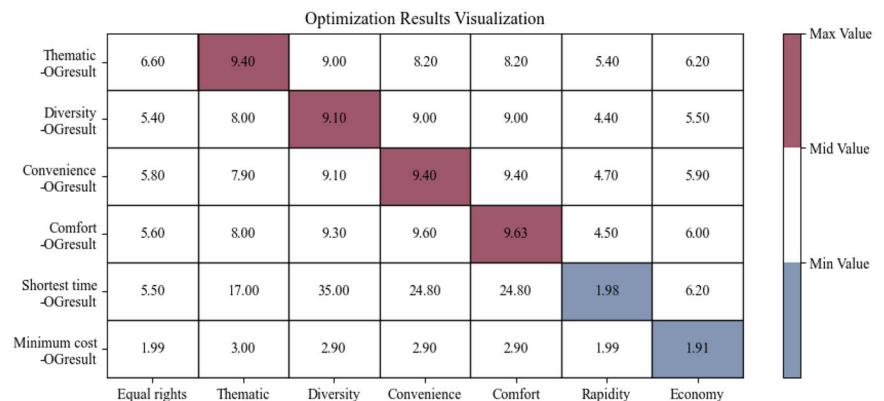


Fig. 13 | Comparison of planning fitness under each Optimization objective.



The convergence performance of the algorithm under different Optimization objectives is presented in Fig. 12, illustrating that the algorithm achieves stable convergence within approximately 100 generations.

Figure 13 displays the results of the improved algorithm for each Optimization objective, where each row corresponds to a distinct Optimization objective and each column shows the fitness results for that objective. The color gradient indicates relative performance within each row: red denotes the maximum value and blue the minimum. The results demonstrate that the algorithm achieves peak performance for certain objectives (e.g., Thematic), while maintaining relatively high fitness across other indicators. This indicates the Optimization algorithm's capability to flexibly adjust indicator values in response to different weight settings, thereby producing high-quality solutions across all Optimization objectives.

Description: Color intensity reflects optimization direction: red indicates maximum values for positive objectives (Thematic/Diversity/Convenience/Comfort), while blue shows the reverse extremes in opposite objectives (Rapidity/Economy).

Trade-off analysis

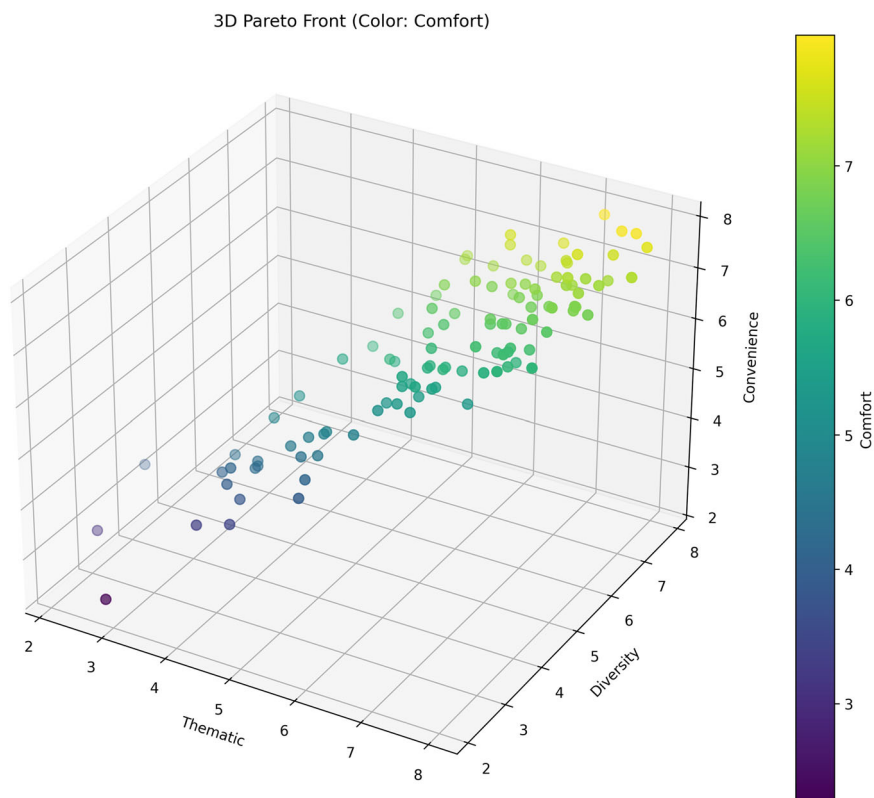
To further analyses the Optimization outcomes, visualizations were produced based on the NSGA-III output. As depicted in Fig. 14, the 3D Pareto front illustrates trade-off relationships among three maximization objectives—Thematic Relevance (TR), Diversity, and Convenience—with Comfort encoded via a color gradient. The visualization reveals a clear upward trend in Comfort scores corresponding with improvements in the other three dimensions, suggesting these objectives tend to be synergistic rather than conflicting within the optimized routing strategy.

Figures 14 and 15 present a systematic analysis of the trade-offs between optimization objectives in cultural route planning. The 3D Pareto front in Fig. 14 demonstrates that solutions with higher thematic coherence (Thematic > 6.5) tend to cluster in regions with elevated Comfort scores, while a clear trade-off emerges between Diversity and Convenience - each unit increase in Diversity beyond 6.0 corresponds to an approximate 15% reduction in Convenience. Figure 15 further elucidates these relationships through its pair plot visualization, revealing strong linear associations between Comfort and both Diversity ($R^2 = 0.68$) and Convenience ($R^2 = 0.71$). The diagonal histograms highlight distinct distribution patterns, with Diversity exhibiting a bimodal distribution (peaks at 3.1 and 6.7) and Comfort showing a right-skewed distribution (mode at 5.2). These visualizations collectively validate the algorithm's capacity to navigate complex objective trade-offs. The spatial clustering of high-performing solutions in the Pareto front, coupled with the quantified relationships between key indicators, demonstrates the framework's effectiveness in balancing competing optimization criteria. The distribution patterns observed in the histograms further reinforce the algorithm's ability to generate diverse solution sets while maintaining consistent performance across multiple dimensions of route quality.

Performance comparison

At the conclusion of the study, three route planning algorithms were compared: the proposed Nsga3ip, Multi-Objective Particle Swarm Optimization (MOPSO), and a Random algorithm. Evaluation metrics included Total Running Time (s), Score-to-Length Ratio, Rational Route Probability, and a composite Optimization Efficiency Index Tables 7, 8.

Fig. 14 | Three-dimensional Pareto Optimal Solutions with Comfort-based Color Mapping (color-mapped by Comfort intensity).



As shown in Table 9, although the Random algorithm has the shortest runtime, its low route quality and rationality make it unsuitable for practical use. Nsga3ip achieves the best overall performance, reducing runtime by 43.7% compared to MOPSO, while improving: ① 53.0% higher solution quality; ② 16.9% greater reliability; ③ 43.5% improved optimization efficiency. These results confirm Nsga3ip's superior balance between computational efficiency and route quality. Figure 16 further illustrates this comparison: (a) shows the average running time of each algorithm; (b) shows the evolution of score-to-length ratios under progressively relaxed time constraints (0–1400 s).

Together, these visuals reinforce Nsga3ip's clear advantage in both optimization performance and practical applicability.

Route visualization and application

In summary, Nsga3ip demonstrates high operational efficiency and Optimization effectiveness, effectively exploring the path solution space to achieve higher quality route planning results. Based on the planning results (Fig. 14), we visualized the tourism trails within the historical city, mapping the primary urban cultural heritage tourist trail (1 → 2 → 3 → 5 → 6 → 7) and designing a guidance system to assist tourists in completing the full trail experience, as shown in Fig. 17 below.

During visits, the system can dynamically adjust the route according to real-time conditions, and time consumption data for the route can be visualized in the Thematic Trail Guiding Interface for tourists' reference.

Discussion

This study proposes a tourism route planning approach based on many-objective Optimization utilizing an improved NSGA-III algorithm, which not only supports customized path design but also exhibits clear advantages over traditional methods. Unlike conventional approaches focusing mainly on shortest distance or lowest cost, the proposed method integrates a wider range of factors—including cultural heritage value, spatial environmental quality, and tourists' subjective experiences—to enhance both cultural depth

and practical feasibility in route planning. By embedding a comprehensive evaluation framework encompassing heritage value and spatial experience, the method incorporates eight indicators related to cultural significance and environmental perception, six reflecting tourist preferences, and incorporates time and cost constraints. The improved NSGA-III algorithm enhances solution diversity and dynamic adaptability, ensuring generated routes are both theoretically robust and practically responsive to real tourist needs. The findings highlight the importance of integrating subjective user experience and cultural context, demonstrating the method's innovation, adaptability, and strong potential for promoting sustainable tourism. Moreover, this approach contributes positively to cultural heritage protection and heritage tourism fields. By quantifying intangible cultural values into planning indicators embedded in algorithmic decision-making, it bridges the gap between cultural interpretation and technological implementation. This not only improves visibility and accessibility of heritage sites but also encourages meaningful visitor engagement, aiding heritage conservation through better spatial management and visitor flow control.

However, limitations remain. First, the completeness and accuracy of the heritage value evaluation system critically influence reliability. Future work should focus on expanding high-quality data sources to enhance precision. Second, while genetic algorithms and many-objective Optimization are theoretically sound, computational efficiency may pose challenges in large-scale or real-time dynamic scenarios. Further refinement of algorithm performance is needed to ensure responsiveness. Third, system implementation and deployment require further validation, especially integration with smartphones and mobile apps, ensuring usability and effective real-world performance. Looking forward, as urban multi-source data accumulates and integrates, route planning algorithms will benefit from richer, more reliable inputs. Future research will focus on developing user-friendly mobile applications integrating public mapping and weather APIs to enable real-time, personalized thematic route recommendations. This will facilitate smarter navigation and contribute to more dynamic, adaptive heritage tourism planning.

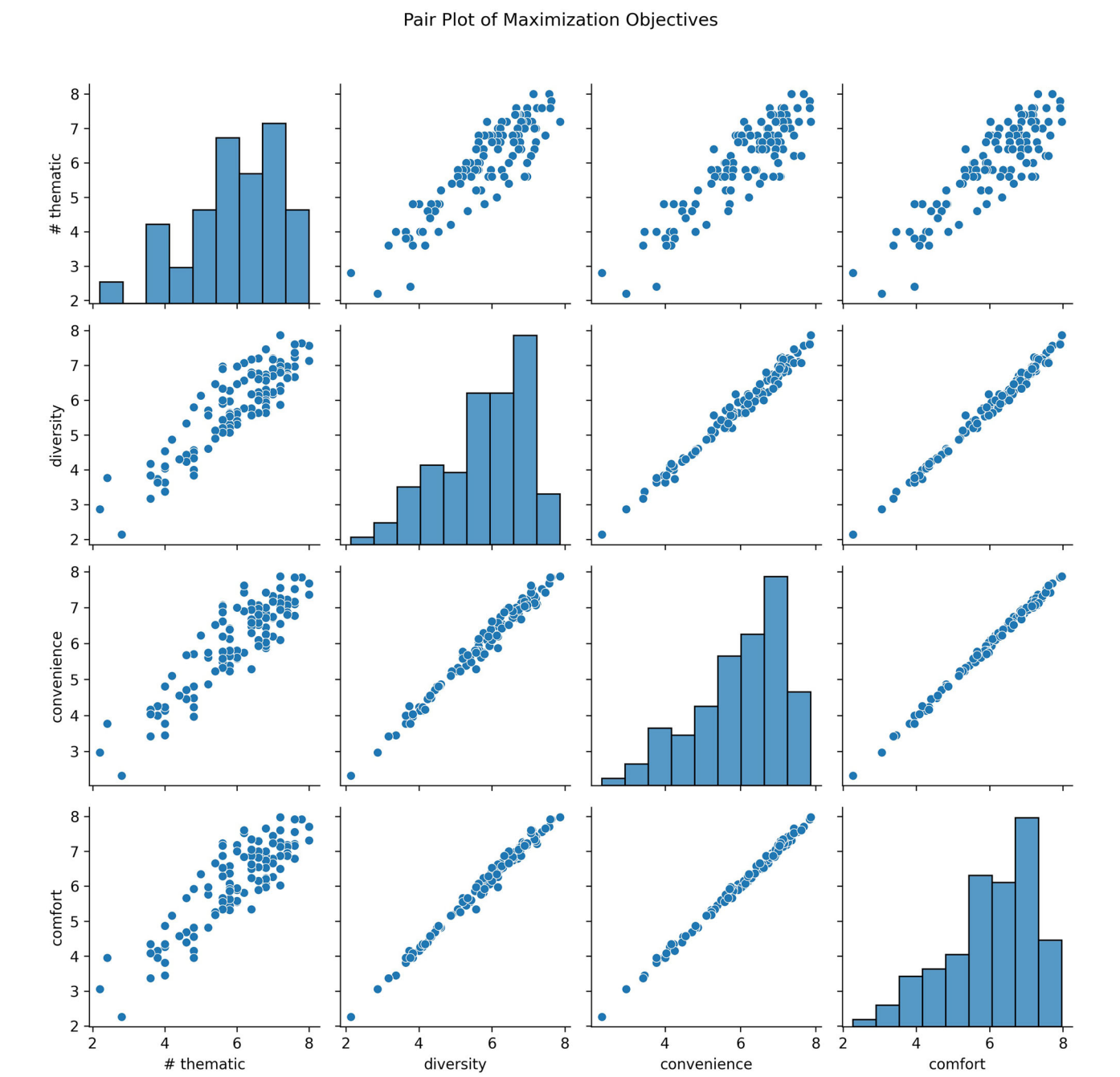


Fig. 15 | Correlation matrix of optimization objectives with marginal distributions.

Table 7 | Initial target weight setting

Optimization Goals	Thematic	Diversity	Convenience	Comfort	Rapidity	Economic
Initial weights	1.0	1.0	1.0	1.0	-1.0	-1.0

Table 8 | Basic parameter settings

GA Variables	Parameters
Population Size	100
Crossover Probability	0.5
Mutation Probability	0.2
Iterations	150

Table 9 | Quantitative metrics of optimization algorithms

Algorithm	Total Running Time (s)	Score-to-Length Ratio	Rational Route Probability	Optimization Efficiency Index ^a
Nsga3ip	19 ± 2.7	4.82 ± 0.31	97%	0.89
MOPSO	32 ± 3.4	3.15 ± 0.28	83%	0.62
Random	7 ± 0.9	1.04 ± 0.17	12%	0.19

^aOptimization Efficiency Index = normalized (Score-to-Length × Rationality) ÷ Time; reflects the overall trade-off between solution quality and computational cost.

Fig. 16 | Temporal evolution of solution quality.
a Average Running Time; **b** Total Score/Path Length Ratio.

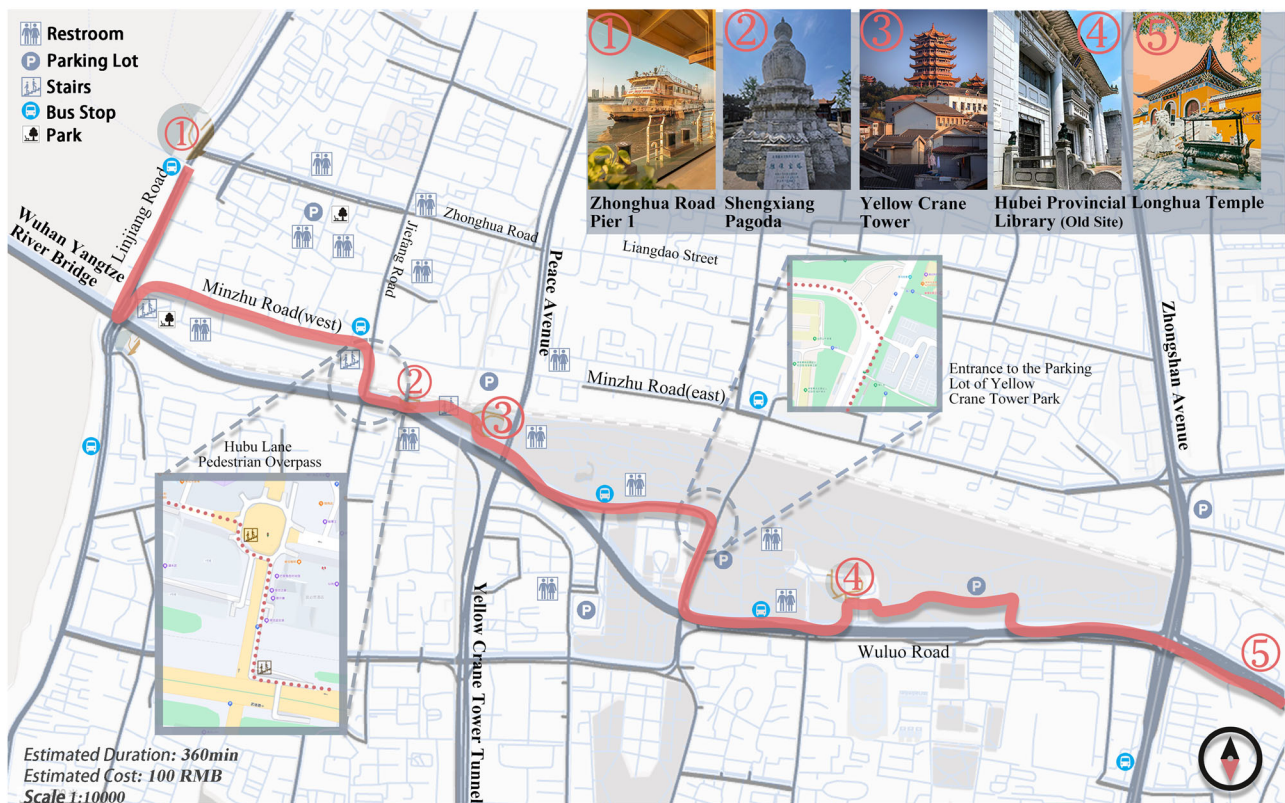
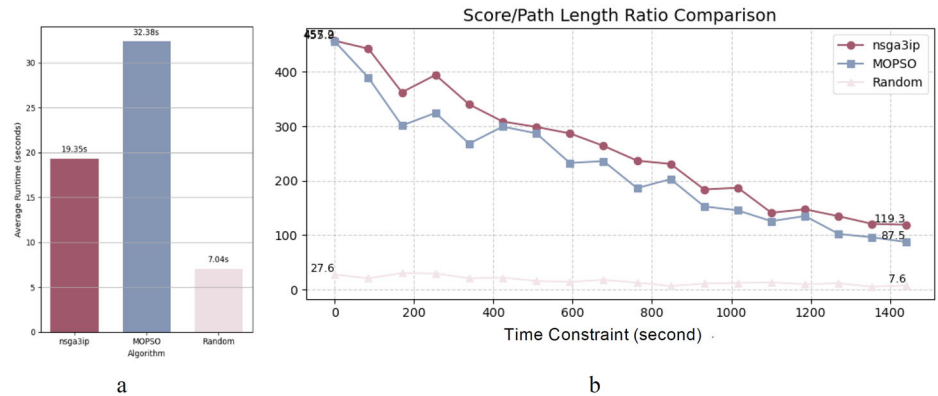


Fig. 17 | Heritage tourism trails in Wuchang Historic District.

Data availability

The datasets and underlying code for this study are not publicly available due to it is being further expanded, but are available from the corresponding author on reasonable request. Moreover, it will be subsequently open-sourced for the research on historical districts conservation and development. The datasets generated and/or analyzed during the current study are available in the [GitHub repository], [<https://github.com/LIM0422/Tourism-Route-Planning-Based-on-NSGA3>] and the supplementary information file.

Code availability

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Competing interests

The authors declare no competing interests.

Additional information

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